

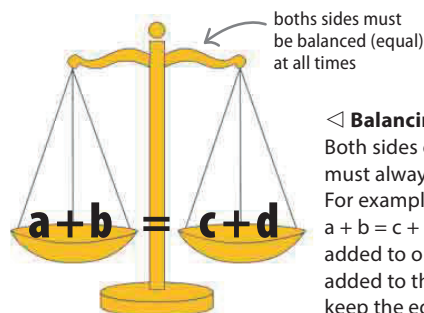
b=? What is algebra?

ALGEBRA IS A BRANCH OF MATHEMATICS IN WHICH LETTERS AND SYMBOLS ARE USED TO REPRESENT NUMBERS AND THE RELATIONSHIPS BETWEEN NUMBERS.

Algebra is widely used in maths, in sciences such as physics, as well as in other areas, such as economics. Formulas for solving a wide range of problems are often given in algebraic form.

Using letters and symbols

Algebra uses letters and symbols. Letters usually represent numbers, and symbols represent operations, such as addition and subtraction. This allows relationships between quantities to be written in a short, generalized way, eliminating the need to give individual specific examples containing actual values. For instance, the volume of a rectangular solid can be written as lwh (which means length $\times$ width $\times$ height), enabling the volume of any cuboid to be found once its dimensions are known.



◁ Balancing

Both sides of an equation must always be balanced. For example, in the equation $a + b = c + d$, if a number is added to one side, it must be added to the other side to keep the equation balanced.

TERM

The parts of an algebraic expression that are separated by symbols for operations, such as $+$ and $-$. A term can be a number, a variable, or a combination of both

OPERATION

A procedure carried out on the terms of an algebraic expression, such as addition, subtraction, multiplication, and division

VARIABLE

An unknown number or quantity represented by a letter

2

+



EXPRESSION

An expression is a statement written in algebraic form, $2 + b$ in the example above. An expression can contain any combination of numbers, letters, and symbols (such as $+$ for addition)

△ Algebraic equation

An equation is a mathematical statement that two things are equal. In this example, the left side ($2 + b$) is equal to the right side (8).